

Prevalence-Calibrated Weighting for Selected Genetic Cohorts

The idea, theory, implementation, and evidential status of `i3pw`

Methodological review for statistical geneticists

11 August 2026

Abstract

Volunteer biobanks and ascertained genetic cohorts need not have the same distribution as their source populations. The `i3pw` package combines weights from a conventional participation model with an exponential tilt chosen to reproduce externally known disease prevalences. In the one-disease case this reduces to the familiar case-control weights K/P for cases and $(1 - K)/(1 - P)$ for controls. In the general case it is an entropy-calibration estimator: useful, computationally modest, and scientifically interpretable, but not a machine for reconstructing unobserved inclusion probabilities from prevalence alone. This report derives the estimator, states the identifying condition, explains its connection to ascertainment correction and label shift, reviews the software, and separates algebraic calibration from empirical evidence. It is written for readers comfortable with likelihoods, generalized linear models, and GWAS, but not necessarily with survey sampling or epidemiologic terminology.

Review scope. This report describes the local version-0.3.1 worktree reviewed on the date above. It evaluates the scientific contract as well as the code structure. Numerical results reported below were regenerated after the simulator corrections and frozen in `report/validation_results.tsv`; the exact local runtime is recorded in `report/validation_environment.txt`. They remain fixed-seed synthetic evidence rather than independent validation.

1 Executive assessment

The central idea is sound and classical in its mathematical form. A register margin is used as a moment constraint; among all participant weights satisfying that constraint, `i3pw` chooses the weights closest in Kullback–Leibler divergence to a user-supplied base weighting. This joins survey calibration [6, 11], entropy balancing [7], density-ratio models [18], and the case-control correction familiar in statistical genetics [15, 13].

Its scientific claim must nevertheless be kept narrow. Exact agreement with an input prevalence is an identity, not validation. The weights equal full-population inverse-probability weights only under a stated density-ratio model. If selection also changes ancestry, severity, genotype, or exposure distributions within disease classes, a marginal prevalence cannot repair those joint distributions.

Table 1: The shortest accurate description of **i3pw**.

Question	Answer
What is computed?	Normalized minimum-divergence weights matching specified population moments.
What information is added?	External prevalences or other population margins, optionally layered on weights from a participation model.
What is identified?	The target population law if its density ratio relative to participants is the base ratio times the chosen exponential tilt; otherwise an information projection, not the true inclusion probabilities.
What is λ ?	The fitted coefficient in the log population-to-participant density ratio. It is not, in general, the coefficient of outcome in a participation logit.
Which IP weight targets the full population?	$1/\pi$, where $\pi = \mathbb{P}(S = 1 \mid X, Y)$. The inverse odds $(1 - \pi)/\pi$ target nonparticipants.
Does matching prevalence validate the method?	No. A held-out margin can diagnose failure, but its standardized mean difference is a heuristic discrepancy, not a formal Sargan–Hansen J -test.
How strong is the present evidence?	Theoretical links, unit tests, and synthetic studies. The package currently has no reported external-cohort validation; its evidential base is simulation-only.

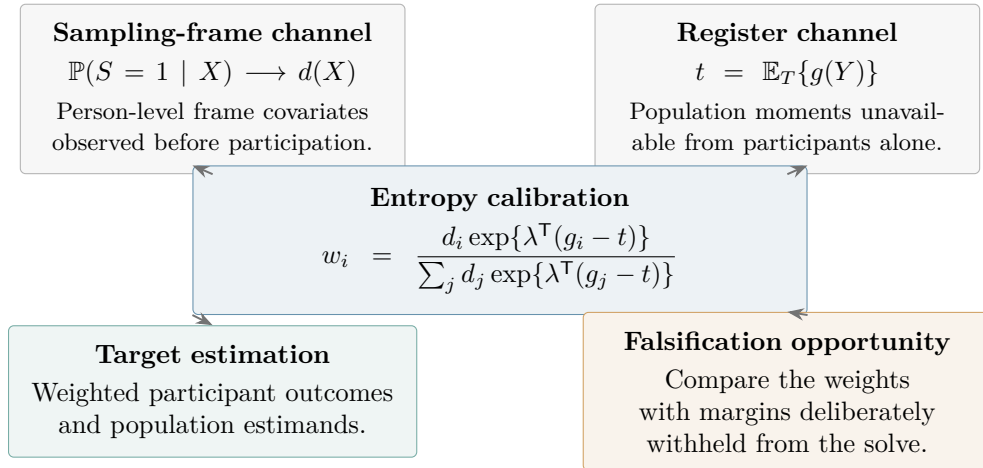


Figure 1: The two information channels and two uses of prevalence-calibrated weighting. The participation model supplies individual variation visible in frame covariates; the register anchors selected population moments. Target estimates use the resulting weights, whereas held-out margins provide a diagnostic that can fail.

2 The problem in a geneticist’s language

Let $S = 1$ denote participation, X variables available on the sampling frame, and Y one or more diagnoses observed in participants. Let T denote the source population to which inference is intended to apply. A volunteer cohort usually samples from $\mathbb{P}(X, Y \mid S = 1)$, whereas the scientific target is a functional of $\mathbb{P}_T(X, Y)$. This is a distribution-shift problem.

For a geneticist, the simplest instance is ordinary case–control ascertainment. If cases constitute P of the analysed sample but K of the population, a population mean is recovered by giving cases relative weight K/P and controls relative weight $(1 - K)/(1 - P)$. `i3pw` keeps that calculation but permits:

1. a nonuniform starting weight $d(X)$;
2. several prevalence, comorbidity, or stratum-specific constraints; and
3. downstream weighted means and liability-scale calculations.

The price of this generality is that one must say exactly what selection model these constraints identify.

2.1 A small epidemiologic dictionary

Table 2: Terms used in this report and their statistical-genetic analogues.

Term	Meaning here	Familiar analogue
Target population	The people about whom the estimand is defined, including nonparticipants.	The population whose disease prevalence K enters a liability transform.
Participation probability	$\pi(x, y) = \mathbb{P}(S = 1 \mid X = x, Y = y)$.	An ascertainment or recruitment probability.
Positivity	Every target-relevant type has nonzero chance to appear.	Both cases and controls, and every required stratum, must be represented.
Calibration	Choose weights so specified weighted sample moments equal known population moments.	Force the weighted case fraction to equal K .
Density ratio	$d\mathbb{P}_T/d\mathbb{P}(\cdot \mid S = 1)$, the weight that transports one law to another.	A sample-to-population prior or ascertainment correction.
Collider bias	An association induced or distorted by conditioning on a common effect such as participation.	Genotype and phenotype both influence volunteering, so analysing volunteers can associate them spuriously [16, 23].

3 Probability setup: what full-population IPW means

Write $Z = (X, Y)$ and

$$\pi(z) = \mathbb{P}(S = 1 \mid Z = z), \quad p_S = \mathbb{P}(S = 1). \quad (1)$$

Bayes' rule gives the population-to-participant density ratio

$$r_T(z) = \frac{d\mathbb{P}_T}{d\mathbb{P}_1}(z) = \frac{p_S}{\pi(z)}, \quad \mathbb{P}_1(\cdot) = \mathbb{P}_T(\cdot \mid S = 1). \quad (2)$$

Consequently, for any integrable target variable $h(Z)$,

$$\mu_T = \mathbb{E}_T\{h(Z)\} = \frac{\mathbb{E}_1\{h(Z)/\pi(Z)\}}{\mathbb{E}_1\{1/\pi(Z)\}}. \quad (3)$$

Equation (3) is the self-normalized, or Hájek, form of Horvitz–Thompson weighting [10, 8]. The unknown sampling fraction p_S cancels. It does not follow that $\pi(z)$ can be learned from participant data and one marginal prevalence: that is the identification problem addressed below.

3.1 Inverse probability is not inverse odds

There are two legitimate transports, but they have different destinations. From Bayes' rule,

$$\frac{d\mathbb{P}_T}{d\mathbb{P}_1}(z) \propto \frac{1}{\pi(z)}, \quad (4)$$

$$\frac{d\mathbb{P}_0}{d\mathbb{P}_1}(z) \propto \frac{1 - \pi(z)}{\pi(z)}, \quad \mathbb{P}_0(\cdot) = \mathbb{P}_T(\cdot \mid S = 0). \quad (5)$$

Thus $1/\pi$ targets the full source population, whereas inverse odds target nonparticipants. Logistic modelling does not make the two targets identical. When participation is rare, $1 - \pi(z) \approx 1$, so the normalized weights can be close; this is an approximation, not an algebraic equivalence.

4 The one-outcome calculation

Let $Y \in \{0, 1\}$, let

$$P = \mathbb{P}(Y = 1 \mid S = 1), \quad K = \mathbb{P}_T(Y = 1), \quad (6)$$

and begin from uniform base weights. An exponential tilt gives cases relative weight $\exp(\lambda)$ and controls weight one. Requiring the weighted case fraction to equal K yields

$$\frac{P \exp(\lambda)}{P \exp(\lambda) + (1 - P)} = K \implies \lambda = \text{logit}(K) - \text{logit}(P). \quad (7)$$

After an irrelevant common normalization, the class weights are

$$d_1 = \frac{K}{P}, \quad d_0 = \frac{1 - K}{1 - P}. \quad (8)$$

For $P = 0.20$ and $K = 0.40$, equation (7) gives $\lambda = 0.9808$; the relative weights are 2.00 for a case and 0.75 for a control. This is worth checking by hand before trusting a larger solve. This is the classical choice-based-sampling result [15] and the core ascertainment factor appearing in liability-scale corrections [13].

The interpretation of λ matters. Let $\pi_y = \mathbb{P}(S = 1 \mid Y = y)$. Equations (2) and (7) imply

$$\lambda = \log\left(\frac{\pi_0}{\pi_1}\right). \quad (9)$$

It is therefore a coefficient in the *population-to-participant density ratio*. Suppose instead that participation obeys $\text{logit}(\pi_y) = \alpha + \gamma y$. Then

$$\lambda = -\gamma + \log \left\{ \frac{1 + \exp(\alpha + \gamma)}{1 + \exp(\alpha)} \right\} = -\gamma + \log \left(\frac{1 - \pi_0}{1 - \pi_1} \right). \quad (10)$$

Only under rare participation is the second term negligible, giving $\lambda \approx -\gamma$. Calling λ the participation-logit coefficient would therefore reverse a sign and omit a non-negligible term in many cohort settings.

5 General entropy calibration

Let $g_i = g(Y_i, X_i) \in \mathbb{R}^k$ be calibration features, with externally known target

$$t = \mathbb{E}_T\{g(Y, X)\}. \quad (11)$$

Examples include diagnosis indicators, pairwise comorbidity indicators, stratum shares, and diagnosis-by-stratum products. Let $d_i \geq 0$ be normalized base weights. Uniform d_i give pure prevalence calibration; estimated $1/\hat{\pi}(X_i)$ supply a covariate correction first.

5.1 Primal and dual problems

For exact calibration, `i3pw` solves the empirical information projection

$$\begin{aligned} \min_{w_1, \dots, w_n} \quad & \sum_{i=1}^n w_i \log \left(\frac{w_i}{d_i} \right) \\ \text{subject to} \quad & w_i \geq 0, \quad \sum_{i=1}^n w_i = 1, \quad \sum_{i=1}^n w_i g_i = t. \end{aligned} \quad (12)$$

This is a Kullback–Leibler calibration estimator [4, 6, 7]. Its dual is

$$\hat{\lambda} = \arg \min_{\lambda \in \mathbb{R}^k} \log \left[\sum_{i: d_i > 0} d_i \exp\{\lambda^\top (g_i - t)\} \right], \quad (13)$$

and the resulting weights are

$$w_i(\lambda) = \frac{d_i \exp\{\lambda^\top (g_i - t)\}}{\sum_{j: d_j > 0} d_j \exp\{\lambda^\top (g_j - t)\}}. \quad (14)$$

Rows with $d_i = 0$ remain outside the support and receive zero weight. The optimization is only k -dimensional even when n is large.

Empirical likelihood and entropy balancing are close relatives, not the same optimization. Both impose moment restrictions and belong to generalized empirical likelihood families. Standard empirical likelihood maximizes $\sum_i \log w_i$ and produces reciprocal-affine weights; the criterion in (12) produces exponential weights. Results from one should not be attributed mechanically to the other [19, 18].

5.2 Ridge stabilization

With ridge parameter $\rho > 0$, the implemented dual becomes

$$\hat{\lambda}_\rho = \arg \min_{\lambda} \left[\log \sum_{i: d_i > 0} d_i \exp\{\lambda^\top (g_i - t)\} + \frac{\rho}{2} \|\lambda\|_2^2 \right]. \quad (15)$$

Its gradient and Hessian are

$$\nabla \ell_\rho(\lambda) = \mathbb{E}_{w(\lambda)}(g) - t + \rho \lambda, \quad (16)$$

$$\nabla^2 \ell_\rho(\lambda) = \text{Cov}_{w(\lambda)}(g) + \rho I_k. \quad (17)$$

At the solution,

$$\mathbb{E}_{w(\hat{\lambda}_\rho)}(g) - t = -\rho \hat{\lambda}_\rho. \quad (18)$$

Ridge calibration is therefore deliberately approximate. It trades bias in the specified margins against less concentrated weights; it must not be reported as exact calibration.

5.3 Feasibility and positivity

For a finite exact solution, t must lie in the relative interior of the convex hull of the observed g_i having positive base weight. A binary target strictly between zero and one is unreachable if the sample contains only cases or only controls. Near a hull boundary, $\|\hat{\lambda}\|$ and the largest weight can explode even when the optimizer reports success. Redundant constraints can also make λ nonunique. These are data-support problems, not numerical inconveniences that a different optimizer can wish away.

6 Identification: when calibration is population weighting

The empirical solve always defines weights if its constraints are feasible. The scientific question is when those weights recover the target law.

Density-ratio assumption. Suppose the true population-to-participant ratio can be represented as

$$r_T(X, Y) = c d(X) \exp\{\theta^\top g(X, Y)\}, \quad (19)$$

where c normalizes the ratio. Here $d(X)$ is the population-to-participant correction already supplied by the base model, and g spans the remaining selection pattern. If the target moments are correct, positivity holds, and the moment map is identifiable, then the population moment equation

$$\frac{\mathbb{E}_1[d(X) \exp\{\theta^\top g\} g]}{\mathbb{E}_1[d(X) \exp\{\theta^\top g\}]} = t \quad (20)$$

identifies θ ; the sample solution estimates it. Under (19), the calibrated weights are proportional to the true full-population IP weights.

This statement is the right meaning of *prevalence informs the weights*. It does not say that prevalence nonparametrically identifies $\pi(X, Y)$. Rather, prevalence identifies the low-dimensional coefficient of a density-ratio family chosen in advance.

6.1 What happens under misspecification

If equation (19) is false, the solve still returns the member of the exponential family that matches t and is closest to the base distribution in KL divergence. Let r_λ denote that fitted ratio. For a downstream variable h , its transport bias is

$$\mathbb{E}_1[\{r_\lambda(Z) - r_T(Z)\}h(Z)]. \quad (21)$$

Calibration gives no general reason for equation (21) to vanish. It does vanish for functions in the affine span of the constrained moments:

$$h(Z) = a + b^\top g(Z) \implies \mathbb{E}_{w_\lambda}(h) = a + b^\top t. \quad (22)$$

This explains both the method's strength and its limit. Anchored quantities are right by construction; transfer to unanchored quantities depends on how well the chosen g captures the relevant distribution shift.

6.2 Label shift and case mix

With uniform base weights, a categorical Y , and $g(Y)$ containing a full set of class indicators, the assumption reduces to

$$\mathbb{P}_T(X | Y) = \mathbb{P}_1(X | Y). \quad (23)$$

That is, only class proportions change. This is prior-probability or label shift [21, 14]. It is plausible for deliberately sampled case-control data when recruitment is random within case status. It is doubtful when, for example, young severe cases enter at a different rate from old mild cases.

For Q comorbid binary diagnoses, matching only the Q marginal prevalences is not full label-shift correction over the 2^Q joint disease patterns. If participation treats a comorbid case differently from the product of its marginal diagnoses, the corresponding joint prevalence or interaction must be constrained.

A known $K = \mathbb{P}_T(Y = 1)$ sets the number of cases, not their composition: $\mathbb{P}_T(X | Y = 1)$ remains unidentified. If population prevalences are available within sex, birth cohort, ancestry, region, or severity strata, interactions such as

$$g_{aq}(X, Y) = \mathbf{1}(A = a)Y_q \quad (24)$$

can represent that structure. The number of constraints then grows quickly, so richer identification consumes positivity and effective sample size. There is no free lunch; the convex hull sends an invoice.

7 Target estimands in statistical genetics

Weighting changes a probability law. Whether that repairs a scientific analysis depends on whether the analysis is a functional of that law and whether the fitted density ratio is correct.

7.1 Participation as a collider

Suppose genotype G or an exposure E influences participation and disease Y also influences participation. Conditioning on volunteers opens a path between the causes of S , potentially

Table 3: What prevalence calibration can and cannot establish for common genetic analyses.

Estimand	When the weighting helps	Principal caveat
Disease prevalence	An exact anchor reports the supplied population prevalence.	This is not an empirical result; uncertainty in the external prevalence remains.
Population mean or absolute risk	Consistent if the fitted ratio transports the participant distribution of the measured variable to the target population.	Marginal disease calibration alone does not fix within-disease selection on severity or prognosis.
Allele frequency or biomarker mean	Can transfer when its relation to the constrained features is stable, or when the full density-ratio model is correct.	A known disease prevalence contains no direct information about genotype or biomarker composition within disease classes.
Logistic GWAS coefficient	Under selection depending on case status alone, the odds-ratio slope is invariant to case-control sampling [17].	When genotype or its causes and phenotype both affect participation, conditioning on $S = 1$ can create collider bias; a marginal K does not identify the required joint correction [16, 23].
Liability-scale R^2 or heritability	Case/control weights recover the population case fraction before the observed-to-liability transform; this parallels classical ascertainment correction [5, 13].	The result inherits the liability-threshold model and the assumed selection law. The current pairwise implementation is quadratic in sample size.
Causal effect	Weights can address selection into the analysed sample.	They do not by themselves address exposure confounding, measurement error, treatment assignment, or interference.

distorting G – Y or E – Y associations. This is not repaired merely by restoring $\mathbb{P}(Y = 1) = K$, because infinitely many joint laws $\mathbb{P}_T(G, Y)$ share the same marginal K . Volunteer-weighting studies in UK Biobank illustrate both the importance and the difficulty of modelling this selection [23, 24, 25].

The distinction from Prentice–Pyke invariance is useful:

- if S depends only on Y , a correctly specified logistic disease slope can remain consistent without prevalence weighting; and
- if S depends jointly on G , X , and Y , population associations require a correction rich enough for that joint selection.

Prevalence calibration can be one component of that correction, but it cannot replace the other components.

7.2 Liability-scale quantities

For population prevalence K , threshold $t_K = \Phi^{-1}(1 - K)$, and $z_K = \phi(t_K)$, a common observed-to-liability conversion is

$$R_L^2 = R_O^2 \frac{K(1 - K)}{z_K^2}. \quad (25)$$

If R_O^2 was estimated in a case-control sample with case fraction P , the Lee et al. ascertainment factor gives

$$R_L^2 = R_{O,\text{asc}}^2 \times \frac{K(1 - K)}{z_K^2} \times \frac{K(1 - K)}{P(1 - P)}. \quad (26)$$

Alternatively, one may first use equation (8) to estimate the population-standardized observed-scale moment and then apply equation (25). Their agreement is expected under pure case/control ascertainment. Neither route protects against recruitment depending on latent liability within case status unless that dependence is modelled.

8 Uncertainty: what is random?

There are at least four uncertainty sources: sampling participants, estimating the base participation model, fitting the tilt, and estimating or transporting the external targets. A standard error that conditions on all weights and targets treats only the first.

8.1 Fixed-weight and calibration-aware linearization

For normalized weights $\tilde{w}_i = w_i / \sum_j w_j$, the fixed-weight variance of $\hat{\mu} = \sum_i \tilde{w}_i h_i$ is estimated by

$$\widehat{\text{Var}}_{\text{fixed}}(\hat{\mu}) = \sum_{i=1}^n \tilde{w}_i^2 (h_i - \hat{\mu})^2. \quad (27)$$

This ignores that the tilt was re-estimated to meet the constraints. The calibration-aware linearization implemented in `i3pw` instead uses

$$\hat{\beta}_\rho = \{\widehat{\text{Cov}}_w(g) + \rho I\}^{-1} \widehat{\text{Cov}}_w(g, h), \quad (28)$$

$$e_i = h_i - \hat{\mu} - \hat{\beta}_\rho^\top (g_i - \bar{g}_w), \quad (29)$$

$$\widehat{\text{Var}}_{\text{cal}}(\hat{\mu}) = \sum_{i=1}^n \tilde{w}_i^2 e_i^2. \quad (30)$$

The software computes equation (28) by penalized least squares rather than explicitly inverting a possibly ill-conditioned covariance matrix.

If $\rho = 0$ and h is an exactly constrained column, the residual in equation (29) is zero up to numerical tolerance. Its conditional sampling standard error is consequently zero: the estimator always prints the fixed target. For $\rho > 0$, equation (18) shows that the target is not fixed, $\hat{\beta}_\rho$ is shrunk, and the anchored margin retains uncertainty. Using the exact-calibration projection after a ridge solve would be wrong.

8.2 External targets, fitted bases, and dependence

Zero conditional standard error is not zero scientific uncertainty. If a registry target $\hat{t}$ has covariance V_t , a first-order combination for an independent external source is

$$\text{Var}(\hat{\mu}) \approx \text{Var}_{\text{participants}}(\hat{\mu} \mid \hat{t}) + D_t V_t D_t^\top, \quad D_t = \frac{\partial \mu}{\partial t^\top}. \quad (31)$$

Temporal mismatch, outcome-definition mismatch, ancestry mismatch, and overlap between registry and cohort create additional bias or covariance that equation (31) does not solve.

The package also offers resampling for its simulation wrapper. A scientifically complete bootstrap should re-fit every estimated stage that would change under repeated sampling, including the base model when applicable. Rare anchors can make bootstrap replicates infeasible; silently discarding precisely those replicates truncates the distribution and narrows intervals.

Family or household relatedness requires cluster-level resampling or a cluster-robust influence calculation. The current simple mean interface does not automate those genetic-cohort complications.

8.3 Augmented IPW

For a participant-only trait V , covariates X available across the population frame, and regression $m(X) = \mathbb{E}(V | X)$, the package implements

$$\hat{\mu}_{\text{AIPW}} = \frac{1}{N} \sum_{i=1}^N \hat{m}(X_i) + \sum_{i:S_i=1} \tilde{w}_i \{V_i - \hat{m}(X_i)\}. \quad (32)$$

In the standard missing-data or data-integration model, this estimator can be consistent when either the outcome regression or the participation weighting is correct [20, 2]; flexible regressions call for sample splitting or cross-fitting [3]. This is not a blanket double-robustness theorem for prevalence calibration. Similarly, the result that entropy balancing is doubly robust in particular causal models [26] does not prove that arbitrary outcome-dependent volunteer selection is repaired by a few prevalence margins.

9 Diagnostics and falsifiability

9.1 Quantities that diagnose the solve

Table 4: Diagnostics, their interpretation, and their limits.

Diagnostic	What it reveals	What it cannot reveal
Optimizer convergence	Whether the numerical dual solve terminated satisfactorily.	Whether the density-ratio family is scientifically correct.
Calibration residual	Whether exact targets were reached, or how far ridge calibration missed them.	Validity: an exact residual is expected by construction.
Kish effective sample size [12]	$\text{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$, a concise measure of concentration.	A universal variance or information measure for every downstream analysis.
Weight range and top-1% mass	Dependence on a small number of observations.	Bias from omitted selection variables.
Support counts	Whether each required class or stratum is represented.	Adequate overlap within unmeasured subgroups.
Held-out SMD	Whether a known margin excluded from the solve is reproduced after weighting.	A formal hypothesis test, or proof that all unobserved margins are balanced.

For a held-out variable a with known population mean t_a , **i3pw** reports

$$\text{SMD}_a = \frac{\sum_i \tilde{w}_i a_i - t_a}{s_{a,\text{sample}}}. \quad (33)$$

The often-used rule $|\text{SMD}| < 0.1$ is a descriptive convention [1]. A held-out failure is valuable: it refutes the proposition that the fitted weighting transports that margin. A small SMD is only partial positive evidence.

Calling equation (33) a Sargan–Hansen J -test would overstate it [22, 9]. A formal overidentification test needs a joint covariance estimate, a quadratic test statistic, degrees of freedom,

and an asymptotic reference distribution while accounting for estimated parameters. The present diagnostic borrows the *logic* of unused restrictions, not that formal test.

9.2 An honest validation design

Known margins should be divided before fitting:

1. use a prespecified subset to calibrate;
2. keep other registry quantities genuinely held out;
3. fit the tilt on one wave or fold and use the stored tilt on a new wave when possible; and
4. evaluate downstream quantities not algebraically determined by the anchors.

Selecting constraints after inspecting held-out failures converts validation into training. Reporting only anchored prevalence error produces impressive zeros and no evidence whatsoever.

10 Software review

10.1 Architecture and public contract

The reviewed source declares version 0.3.1 and an alpha development status. Its design has a sensible separation between a real-cohort interface and simulation machinery.

10.2 Scientific contracts that must remain explicit

10.3 Computational properties

Let n be the number of participants with positive base weight, k the number of constraints, p the number of frame covariates, I optimizer iterations, and B bootstrap replicates.

For the core calibration, k is normally small and the implementation is reasonable. The main practical failure is statistical rather than computational: adding many sparse stratum-by-disease constraints can exhaust support before it exhausts RAM. The liability module is the exception; materializing n^2 pairwise similarities is untenable at modern biobank scale.

10.4 Remaining release gaps

Table 5: Main software components in the reviewed **i3pw** worktree.

Component	Public role	Review
<code>calibrate</code> and <code>CalibrationFit</code>	Real-cohort entry point using participant outcomes, external targets, optional base weights, strata, joint moments, and held-out margins.	The correct conceptual front door; it avoids requiring simulated ground truth.
<code>entropy_balance</code> and <code>apply_tilt</code>	Core dual solve and transfer of a fitted tilt to new rows.	Small convex problem, log-scale stabilization, explicit diagnostics, and preservation of zero-weight support.
<code>lasso_propensity</code> and <code>lasso_ipw</code>	Covariate-only participation baseline.	Useful comparator, but a real application needs frame-level participation labels and compatible covariates for participants and nonparticipants.
<code>calibration_ipw</code> and <code>monte_carlo</code>	Simulation wrappers with known population truth.	Appropriate for method development, not the interface to advertise as real-data inference.
<code>balance</code> and <code>uncertainty</code>	Held-out discrepancies, effective sample size, linearization, bootstrap, and target sensitivity.	Good instincts; target uncertainty, clustering, and a formal specification test remain user responsibilities.
<code>aipw</code>	Outcome-regression augmentation for a participant-only trait when X is known for the population frame.	Cross-fitting is available; robustness claims still require the usual missing-data conditions.
<code>liability</code>	Threshold-model simulations and weighted pairwise moment estimators.	Scientifically relevant, but the explicit $n \times n$ similarity matrix does not scale to biobank sample sizes.

Table 6: High-impact semantic and numerical contracts in the corrected design.

Issue	Correct contract	Consequence
Meaning of the tilt	λ belongs to the log population-to-participant density ratio, as in equations (9)–(10).	Do not report it as the outcome coefficient in a participation logit.
Target of base weights	$1/\hat{\pi}$ transports participants to the full population; $(1 - \hat{\pi})/\hat{\pi}$ transports them to nonparticipants.	The inverse scheme is the full-population default; an odds option needs an explicit different target.
Simulation oracle	A full-population oracle is the complete simulated test-fold mean, unavailable in real data.	An inverse-odds mixture is not a full-population oracle and should not be labelled one.
Ridge calibration	The achieved residual equals $-\rho\hat{\lambda}_\rho$, and the influence projection contains the same ρ .	Only exact $\rho = 0$ anchors have zero conditional standard error.
Zero base weights	Zero-weight rows are excluded from the exponential shift and remain zero.	An extreme feature value outside the positive support must not destabilize all weights.
Held-out balance	The SMD is a heuristic discrepancy using an unused margin.	It may be called a falsification diagnostic, not a formal J -test.
Synthetic selection	The generator uses Bernoulli logistic participation with coefficients chosen so expected sample count and outcome totals meet configured targets.	Realized sample size and prevalences are random; the mechanism is not fixed-size probability-proportional-to-size sampling.

Table 7: Leading computational costs. These are algorithmic orders, not timing benchmarks.

Operation	Approximate time	Approximate memory	Comment
Entropy dual	$O(Ink)$	$O(nk)$ for features	L-BFGS avoids forming a dense $k \times k$ Hessian.
Stratified design	$O(nk)$	$O(nk)$	k can be roughly strata times outcomes, plus share and interaction constraints.
LASSO participation model	Solver- and CV-dependent	$O(np)$ before expansion	Pairwise interactions increase the design toward $O(np^2)$.
Calibration bootstrap	$O(BInk)$ plus optional base refits	One replicate plus stored output	Parallelization is natural, but rare anchors can cause infeasible replicates.
Liability pairwise moment	$O(n^2p)$ to form XX^T	$O(n^2)$	This is the principal barrier to biobank-scale use. Blocked matrix products or summary-statistic identities are needed.
Synthetic data generator	Repeated $O(Nq)$ selection objectives, plus covariance work	$O(Np + Nq + p^2)$	Useful for controlled experiments; not normally a production bottleneck.

Table 8: Important gaps that remain after correcting the scientific semantics.

Gap	Consequence	Recommended next step
Real-data resampling interface	The bootstrap and prevalence-sensitivity helpers take the simulation Dataset ; users of calibrate must assemble equivalent refits themselves.	Provide a callback- or array-based bootstrap that refits the base, tilt, and estimand without simulated ground truth.
External-target and cluster variance	CalibrationFit.mean treats external targets as fixed and independent units as the sampling units.	Accept target covariance and cluster identifiers, or document a supported cluster-bootstrap recipe.
Liability scaling	The explicit $n \times n$ similarity matrix is incompatible with biobank-scale n .	Use blocked accumulations, randomized trace methods, or algebraic sufficient statistics that avoid storing pairwise matrices.
Rendered documentation	The corrected distributions include the Markdown theory, examples, LaTeX report, and citation record, but there is no rendered versioned site or link checker.	Publish versioned HTML/PDF documentation and check external and internal links in CI.
Frozen environment and automated evidence	A citation file, structured post-correction results, and a local environment manifest are now present, and CI smoke-tests one packaged example; there is no cross-platform dependency lock, replicated-validation job, or comparison of outputs with the frozen table.	Record a locked environment and add a reproducible benchmark job with explicit tolerances and retained logs.

11 Evidence review

The mathematical ingredients of `i3pw` are well established. What is new is their particular packaging and proposed use of register prevalences to augment participation weights. Evidence for that applied proposal must be judged separately from evidence for entropy calibration itself.

Table 9: Current evidence and what each layer can support.

Evidence layer	What it supports	What it does not support
Established literature	IPW, survey calibration, information projection, density-ratio tilting, and case-control ascertainment correction.	That the selected g spans selection in a particular biobank.
Unit tests	Array contracts, numerical convergence, known special cases, support checks, and selected uncertainty calculations.	External validity, realistic misspecification, or calibrated interval coverage in a real cohort.
Synthetic data generator	Controlled comparison when outcomes and Bernoulli-logistic participation are known.	Performance under an unknown real recruitment process.
Example scripts and frozen local table	Illustrations of pure outcome selection, covariate-plus-outcome selection, collider settings, comorbidity, and liability models; selected corrected runs are recorded in <code>report/validation_results.tsv</code> .	Prespecified, independently replicated benchmark evidence.
Real-cohort study	None is included in the present evidence base.	No empirical claim about bias reduction in iPSYCH, UK Biobank, or another cohort follows from the repository alone.

11.1 Fresh post-correction smoke evidence

Table 10 records the most informative results from a fresh local run on 11 August 2026 using Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1, and scikit-learn 1.9.0 with the corrected `i3pw` 0.3.0 worktree. Replicated scripts use seeds $0, \dots, R-1$. These are regression and plausibility checks, not a substitute for an independently designed validation study. Figure 2 visualizes the three contrasts that are most useful for interpreting those checks.

Table 10: Selected post-correction synthetic results. Values are mean $\pm$ standard deviation across repetitions unless stated otherwise.

Experiment	Quantity	Result	Interpretation
honest_benchmark , no X channel, $R = 12$	Absolute percentage error on an unanchored outcome	Naive 80.78 ± 6.66 ; calibrated LASSO base 81.61 ± 6.03 ; calibrated uniform base 82.24 ± 5.96 .	Anchoring one disease did not transfer to another.
honest_benchmark , X channel strength 1.5, $R = 12$	Worst held-out SMD ; absolute error in held-out X_0 mean	LASSO base 0.209 ± 0.079 , 0.126 ± 0.072 ; uniform base 0.482 ± 0.238 , 0.317 ± 0.211 .	When a covariate channel exists, the base participation model materially helps.
doubly_robust_trait , $R = 20$	Mean bias; mean absolute bias for a participant-only trait	Naive $-0.0905, 0.1008$; LASSO IPW $-0.0252, 0.0636$; calibration $0.0537, 0.0864$; AIPW $-0.0009, 0.0584$.	Augmentation performed best in this particular, outcome-model-friendly design.
monte_carlo , $R = 20$	Anchored prevalence error; calibration ESS	Calibration error 0.00 ± 0.00 ; ESS 153 ± 19 .	The zero error is the constraint, not validation; ESS quantifies its variance cost.

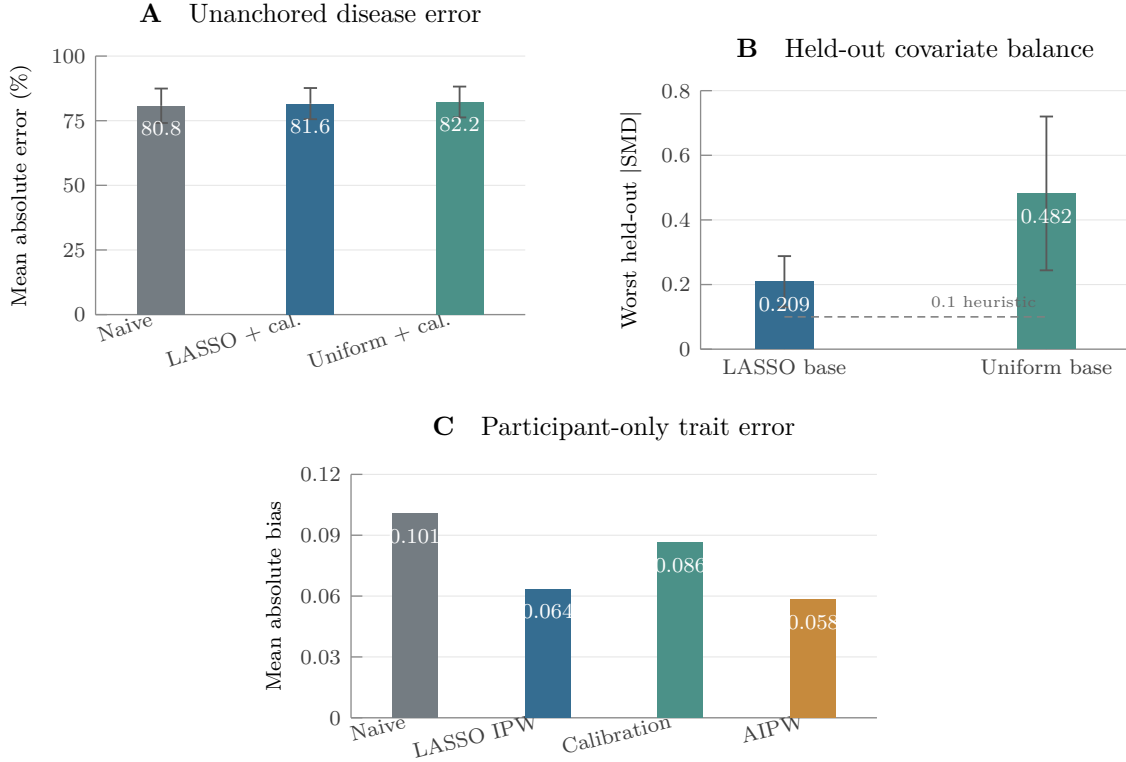


Figure 2: Three fixed-seed synthetic contrasts from the frozen validation artifact. Panel A shows mean absolute percentage error for an outcome excluded from calibration when selection has no direct covariate channel ($R = 12$). Panel B compares base weights when covariates do affect selection ($R = 12$); the dashed 0.1 line is only an SMD rule of thumb, not a hypothesis test. Panel C shows mean absolute bias for a participant-only trait in one outcome-model-friendly design ($R = 20$); it is not a general method ranking. Whiskers in A and B are ± 1 across-seed standard deviation. The frozen artifact does not retain an uncertainty summary for the mean absolute biases in C.

The table and figure support the package’s deliberately narrow messages: exact anchors are tautologies, cross-outcome transfer is weak, and the covariate base matters when the data-generating process actually contains a covariate selection channel. They do not establish bias correction in a real cohort, interval coverage, or robustness to an unknown selection law. For stronger evidence:

- error on a constrained prevalence must be identified as a match by construction;
- at least one outcome, stratum, or population margin must remain unanchored;
- simulations must vary both covariate-driven and outcome-driven selection;
- misspecified tilts, target error, weak support, and within-case selection must be included; and
- empirical validation needs a population quantity unavailable to the fitting procedure, preferably in a later cohort wave or linked register.

11.2 Recommended validation matrix

Table 11: Minimum simulation and empirical validation matrix for a methods paper.

Axis	Levels	Primary outputs
Selection law	X -only; Y -only; additive $X + Y$; $X \times Y$; within-case severity; genotype-plus-outcome.	Bias, RMSE, interval coverage, ESS, maximum weight.
Anchor information	Correct pooled prevalence; stratum-specific prevalence; joint comorbidity; perturbed targets; partial anchors.	Anchor residual and transfer error on held-out outcomes and margins.
Support	Common, rare, and extremely rare disease; sparse ancestry/sex/age cells.	Failure probability, convergence, weight concentration, bootstrap discard rate.
Base model	Correct, misspecified, regularized, cross-fitted, and absent.	Incremental value of the base versus the prevalence tilt.
Comparator	Unweighted, covariate IPW, exact calibration, ridge calibration, oracle full-population law in simulation.	Bias–variance trade-off on the same target estimand.
Real data	Prespecified fit margins and separate held-out register margins or recruitment wave.	External discrepancies with uncertainty, not merely anchored matches.

12 Practical analysis protocol

A defensible analysis can be organized in nine steps.

1. **Define T .** State geography, calendar period, age range, ancestry scope, survival condition, and disease definition. A prevalence from a different population is not a magic number; it is a different estimand.

2. **Define the target quantity.** A prevalence, population mean, regression coefficient, and liability-scale variance component need different estimating equations. Weighting is not the estimand.
3. **Build the base correction.** Fit participation using variables observed compatibly in the cohort and its sampling frame. For the full population use $1/\hat{\pi}(X)$, not inverse odds. Separate training and analysis folds when a flexible model is used.
4. **Prespecify g .** Begin with scientifically justified prevalence margins. Add strata or interactions only when population targets and participant support exist. Reserve some credible margins for validation.
5. **Fit and inspect.** Report convergence, residuals, tilt coefficients, support counts, ESS, range, quantiles, and top-weight mass. For ridge, report ρ and achieved rather than nominal margins.
6. **Try to refute the weighting.** Evaluate held-out margins and a transferred tilt on a later fold or wave. Do not call an SMD threshold a p -value.
7. **Estimate uncertainty at the right level.** Use the penalized calibration influence adjustment corresponding to the fitted ρ , or re-fit the entire pipeline in a bootstrap. Respect family clusters and quantify uncertainty in external targets.
8. **Stress the assumptions.** Perturb prevalence, alter trimming or ridge, coarsen strata, and examine plausible within-case selection. Trimming after exact calibration breaks the constraint and must be followed by reporting the new residual.
9. **Report weighted and unweighted analyses.** Large changes deserve explanation; small changes can reflect either little bias or an uninformative weight. Neither is established by the point estimate alone.

Table 12: Items that should accompany a published i3pw analysis.

Item	Minimum disclosure
Target population	Eligibility, time, place, ancestry handling, outcome definition, and source of every external margin.
Base model	Training frame, predictors, missing-data handling, validation, clipping, and whether weights are $1/\pi$ or inverse odds.
Calibration model	All g columns and targets, exact or ridge objective, ρ , strata, joint moments, and any post-fit trimming.
Support and concentration	Cases and controls per cell, ESS, weight quantiles, maximum, and top-1% mass.
Validation	Which margins were held out before fitting, their discrepancies, and whether the tilt was transferred to independent data.
Inference	Influence or bootstrap procedure, refitted stages, clustering unit, failed replicates, and treatment of external-target uncertainty.
Sensitivity	Plausible target perturbations and departures from within-class exchangeability.
Reproducibility	Software version or commit, random seeds, environment, frozen tables, and code that recreates every reported number.

13 Overall judgment

`i3pw` is best described as *prevalence-calibrated density-ratio weighting*. That description is more precise than the expansion of the package name. The package does not infer unrestricted individual participation probabilities. It fits the smallest exponential modification of base weights that matches trusted population moments, and it becomes true full-population IPW when the density-ratio assumption in equation (19) is correct.

For statistical genetics, the method is most persuasive as a generalization of case-control ascertainment correction:

- it naturally combines a frame-based participation model with known register prevalences;
- it accommodates several diagnoses, joint moments, and stratum-specific margins; and
- it makes the correction inspectable through the tilt and weight diagnostics.

The risks are clear: marginal prevalences cannot identify a joint distribution; exact anchors are tautologies; weak support creates extreme weights; and validation remains synthetic. The next milestone is a prespecified external-cohort or held-out-register study of quantities withheld from fitting.

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